Post-tuberculosis lung disease

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Shareable abstract (@ERSpublications)
TB is the leading global cause of death from a single infectious agent, despite that it can be prevented and treated. TB-associated lung diseases increase morbidity during and after a TB episode. More can be done to lessen the burden on affected persons. https://bit.ly/3WJrmNQ

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Abstract

Tuberculosis (TB) burden concentrates in low-income settings and remains the leading global cause of death from a single infectious agent, despite that it is preventable and treatable. TB-associated lung diseases (TBALD), a broad range of respiratory abnormalities which can start before or during a TB episode, may increase morbidity. TBALD may persist after successful completion of TB treatment as post-TB lung disease (PTLD). PTLD varies in severity and is characterised by persistent respiratory symptoms and lung impairment that can significantly impact social activities, health-related quality of life, and long-term survival. Risk factors for PTLD include increasing age, smoking, HIV infection, delayed diagnosis, and poor socioeconomic conditions.

Action to limit PTLD may be taken before TB develops through TB screening, early diagnosis and TB preventive treatment, during treatment of TB, and upon its completion. Early detection, clinical assessment, and tailored management (including smoking cessation, immunisation, addressing respiratory comorbidities, pulmonary rehabilitation and social protection) can mitigate impairment and disability. Healthcare providers and national programmes play a vital role through clinical follow-up, patient education, and integration of TBALD care into broader health and social protection services. Sustained funding and research are crucial for this and to develop new tools to enhance care.

Tuberculosis in the world: its burden and significance to global health today

Much of the disease, death, disability, hardship and financial loss wrought by tuberculosis (TB) across the world is preventable or treatable with approaches and tools that are widely available, affordable, and highly cost-effective. Despite this, in 2023, TB caused an estimated 1.25 million deaths, making it the leading cause of death from a single infectious agent [1]. Approximately 11 million people continue to fall ill with TB every year and TB incidence has risen following the disruptions brought about by the coronavirus disease 2019 pandemic. The global TB burden is unevenly distributed, and most people with TB are concentrated in Asian and African countries (figure 1). Urgent action is required to end the global TB

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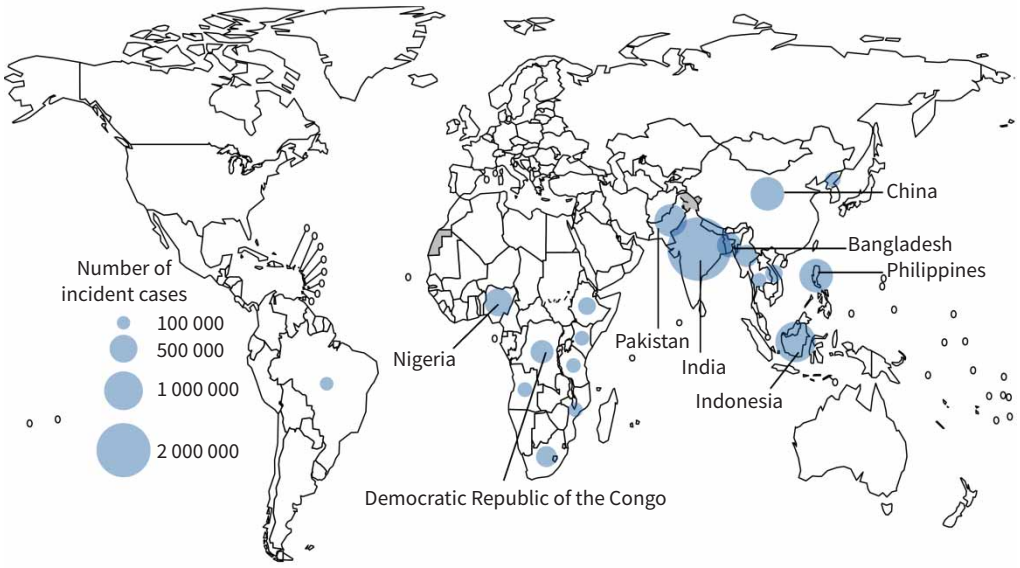


FIGURE 1 Estimated number of incident tuberculosis cases in 2023, for countries with ⩾100 000 incident cases. Reproduced from [1] with permission.

epidemic, and goals have been adopted by all member states of the United Nations and the World Health Organization (WHO) to achieve this [2].

TB is caused by *Mycobacterium tuberculosis*, a bacillus which is expelled into the air by people who have TB disease (*e.g.* by coughing) and can infect others. Approximately a quarter of the global population is estimated to have been infected with *M. tuberculosis* [3]. Following infection, a small proportion of those infected will develop TB disease sometime during their lifetime. The risk of an individual developing TB is influenced by social, demographic and health-related factors. Poverty predisposes to a host of factors that increase a person's risk of being infected with *M. tuberculosis*, progressing from infection to disease and sustaining unfavourable health and social outcomes once disease occurs and even after TB treatment completion. Thus, relief of poverty is at the heart of public health approaches to reduce TB, by mitigating overcrowding, indoor and outdoor air pollution, malnutrition, and risks to health such as substance use and smoking, as well as improving access to preventive and curative healthcare.

TB typically affects the lungs, but it can involve other sites in the body as well. People who develop TB disease may feel sick, but many only have subtle or nonspecific clinical features like cough, chest pain and fatigue, which can sometimes persist for years. Nationally representative prevalence surveys have shown that approximately half of people with bacteriologically confirmed TB do not report symptoms [4, 5]. People with asymptomatic or subclinical TB may still endure damage to the lungs and transmit *M. tuberculosis* to others. Medical measures to reduce TB transmission such as screening, active TB case finding and provision of TB preventive treatment as well as nutritional interventions and social protection measures can reduce the occurrence of TB. Early detection combined with prompt treatment avert unnecessary death, disability and long-term consequences.

TB has a propensity to interact with other lung conditions and cause lasting impairment to the lungs or other organs that it affects. TB-associated lung diseases (TBALD) and post-TB lung disease (PTLD) are topical subjects of discussion in the space of TB care (box 1) [6, 7]. In this review we summarise current knowledge in a rapidly evolving field and discuss ways to alleviate comorbidities and health risks that can lead to respiratory damage in people with TB. We focus on the preservation of lung health once TB is diagnosed.

BOX 1 Tuberculosis (TB)-associated lung diseases and post-TB lung disease

- In this article, TB-associated lung diseases (TBALD) refer to a broad range of respiratory abnormalities, with or without symptoms, concurrent or attributable at least in part to TB, that can start before or during a TB episode, and can persist long after treatment completion.
- Post-TB lung disease (PTLD) is a form of TBALD that occurs after successful completion of TB treatment.

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TB-associated lung diseases

The term "TB associated" acknowledges that the cause of lung diseases in people with TB is often multifactorial and is not always fully attributable to TB. These include a broad array of respiratory pathologies such as pulmonary fibrosis, bronchiectasis, aspergillosis and cancer. Recognition of TBALD signals the importance of assessing lung function and managing lung diseases alongside the institution of TB antimicrobial therapy. Successful completion of TB treatment does not stop the TB disease process from damaging the lungs permanently [8]. The host *versus* mycobacterium immune response results in chronic inflammation, increased mucosal secretions in the respiratory tract, raised systemic inflammatory cytokines and destruction of lung parenchyma, bronchial vasculature, airways and other vital structures of the respiratory system [9]. This compromises respiratory function in ~60% of TB survivors, with chronic symptoms including cough and breathlessness, impairing health-related quality of life, contributing to disability and reducing long-term survival [10–13]. Children and adults who survive TB can also have chronic respiratory diseases which are commonly found in the general population, as many of the settings with a high burden of TB also concentrate other risk factors for poor lung health such as undernutrition, indoor and outdoor air pollution and recurrent respiratory viral and bacterial infections.

People on TB treatment experience symptoms and ill health that cannot be addressed by TB medications alone. It is important that TB treatment providers consider symptoms, measures of lung health and other patient-reported health outcomes when providing a person-centred approach to TB treatment, from the moment a person is diagnosed with TB. This is consistent with universal principles of human rights and health agreed by the United Nations and a core principle of WHO's End TB strategy [14, 15]. Symptoms and measurable abnormalities of lung function contribute to significant TB-related morbidity, estimated to be 122 million disability-adjusted life years (DALY) worldwide, consisting of 64 million DALY for TB disease alone and 58 million DALY occurring after TB treatment is completed [12].

Preventing and treating TB are likely to be the most important interventions to mitigate TBALD and PTLD. Regimens to prevent and treat drug-susceptible and drug-resistant TB are widely accessible today, including shorter treatment options [16, 17]. However, data on TBALD and lung health measures other than medication are rarely routinely collected or studied in TB treatment trials, missing opportunities to inform policy decisions about TB treatment options that could improve outcomes for TB survivors [18].

Post-TB lung disease

The clinical spectrum of PTLD varies widely, but typically includes respiratory signs and symptoms, decreased lung function and systemic effects, and an increased risk of death (figure 2). It can present as a subtle asymptomatic radiological abnormality or with debilitating respiratory symptoms such as haemoptysis and breathlessness causing hypoxia, reduced lung function and obvious radiological abnormality with extensive lung destruction with cavities, bronchiectasis and fibrotic contraction [19, 20]. A DELPHI consensus process during an international PTLD symposium defined the condition as "evidence of chronic respiratory abnormality, with or without symptoms, attributable at least in part to previous tuberculosis" [21]. PTLD can manifest after successful completion of TB treatment and microbiological cure as abnormal lung function (commonly detected by spirometry), persistent or progressive radiological abnormalities, or chronic respiratory symptoms, alone or in combination. In addition to PTLD, post-TB impairment can also affect the cardiovascular, neurological, psychological and

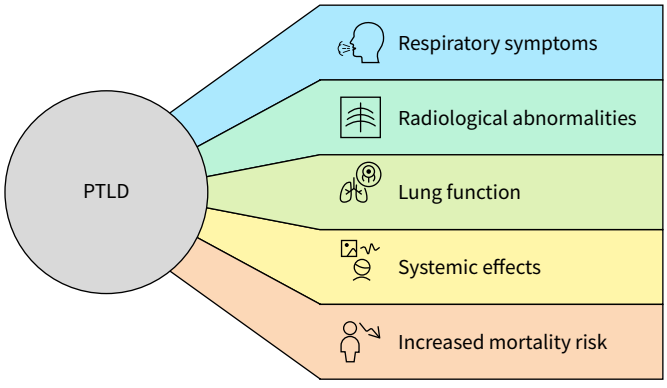


FIGURE 2 Spectrum and components of post-tuberculosis lung disease (PTLD).

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musculoskeletal systems [22]. There is also evidence that TB survivors have an increased risk of lung cancer after TB [10, 23]. It is perhaps not surprising therefore, that increasing evidence from around the world demonstrates increased mortality among TB survivors despite previously being treated successfully for TB with microbiological cure [11, 24].

Clinical assessment with spirometry and radiological imaging are helpful to evaluate PTLD.

Spirometry abnormality

Spirometry abnormality following TB treatment completion is strikingly frequent. In some studies, the majority of patients have abnormal lung function at the end of treatment [10, 25]. The pattern of abnormality may be obstructive, restrictive, or mixed. Measurement of lung volumes adds important additional information and allows the confirmation of restriction, particularly in those with fibrosis and pleural disease. A significant reduction in diffusion capacity of the lung for carbon monoxide (D_{LCO} or T_{LCO}) that is out of keeping with spirometry may indicate the presence of pulmonary hypertension, but also structural lung disease. While clinically useful, measurement of lung volumes requires resources that may not be available in many settings. Another consideration, particularly early in TB treatment, prior to sputum culture conversion, is aerosol generation during lung function testing and the potential for transmission of TB. Lung function testing during TB treatment may help assess response to TB treatment, or if TBALD is present. The addition of host-directed therapies or the optimisation of, or choice of TB treatment regimen, may also affect the presence or severity of PTLD.

Radiological changes

The radiological status of the lungs at TB treatment completion reflects the residual damage attributable to TB, as well as any pre-existing lung disease or respiratory insult such as radiological emphysema from tobacco smoke exposure, asthma or prior respiratory tract infections such as severe pneumonia. Sometimes, intercurrent respiratory illness or TBALD including consolidation from bacterial pneumonia, nodular infiltrates or cavities from fungal or non-tuberculous mycobacterial disease or bronchiectasis can also be detected by radiological imaging.

Chest radiography is a vital tool for the diagnostic workup of TB disease and is also used to assess a person's response to TB treatment. Radiological changes attributable to TBALD and PTLD can also be assessed by chest radiography and, if available, computed tomography (CT) scanning [26]. The pattern of radiological abnormality in PTLD varies widely in both chest radiographs and CT scans. While the reporting for PTLD by either chest radiography or CT scanning is not standardised, common radiological findings of PTLD include cavitation, bronchiectasis, and pulmonary fibrosis, typically in the upper lobes [10]. CT also captures pulmonary nodules, consolidation, emphysema, pleural thickening and can be used to assess the pulmonary artery relative to the aorta to reflect chronic changes associated with (but not diagnostic of) pulmonary hypertension, adding invaluable information to inform the management of PTLD [26–28]. Positron emission tomography (PET) has also been used to provide valuable insights into the metabolic activity of intraparenchymal pulmonary lesions as well as lymph nodes prior to, during and after TB treatment and reflects both microbiological and TB injury related inflammation, and less commonly, malignancy risk of a metabolically active, suspicious lung nodule (case example in box 2). Decisions to undertake CT and PET scanning should balance the usefulness of any additional information they can provide to enhance care, the clinical context, risk factors, and chest radiography findings *versus* its availability, cost and additional radiation exposure. The relationship between radiological changes and symptoms or quality of life of TB survivors is not yet well understood. It is expected that the more severe the abnormalities on imaging, the greater the symptom burden, particularly when much of the normal lung is destroyed or significantly damaged. However, further research is needed to better understand these associations.

Respiratory symptoms and quality of life

Patients with PTLD may experience persistent cough, sputum production, breathlessness, chest pain, fatigue or reduced 6-min walk distance (6MWD) despite successful treatment for TB. In a meta-analysis of 13 studies with 3179 participants, 73% of patients had a modified Medical Research Council (mMRC) dyspnoea score of 1 or 2 (dyspnoea on strenuous exercise or walking up a slight hill) [10]. Alarmingly, one in four participants reported walking slower than most people, needing to stop because of breathlessness after a few minutes of walking on a flat surface or being unable to leave their house due to dyspnoea (mMRC score of 3–5). Patients who had completed TB treatment had 6MWD only ~80% of what was predicted for their age and sex, with a pooled mean of 440 m. One review found significantly decreased lung function in people recovering from pulmonary TB when compared with controls, with a mixed obstructive and restrictive picture with predominantly airflow obstruction [29]. The TRUNCATE TB trial,

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BOX 2 Post-tuberculosis (TB) lung disease (PTLD) with COPD and synchronous lung cancer

A 60-year-old foreign-born man was referred for assessment following an abnormal chest radiography detected at immigration. He gave a history of two previous TB treatment episodes treated in his country of origin >2 years previously. No microbiology results from the time were available.

Following clinical assessment, examination of spontaneous and induced sputa and computed tomography (CT) imaging (figure 3), a diagnosis of rifampicin-resistant TB was made (rpoB gene mutation detected in *M. tuberculosis* strains on GeneXpert MTB/RIF assay). A course of bedaquiline, pretomanid, linezolid and moxifloxacin ("BPaLM" regimen) for a planned duration of 6 months was started. Baseline and repeated sputum cultures were negative on final mycobacterial culture at 8 weeks.

While undergoing antimicrobial treatment, TB-associated lung disease (TBALD) was suspected and lung function testing undertaken. Resting lung function demonstrated mild airflow obstruction (Global Initiative for Chronic Obstructive Lung Disease criteria) with a significant bronchodilator response, mild gas trapping and borderline hyperinflation of static lung volumes. There was a moderate reduction to diffusion capacity when adjusted for alveolar volume. The forced expiratory volume in 1 s (FEV_1) improved from 3.07 L (81%) to 3.85 L (102%) and the forced vital capacity (FVC) from 6.31 L (130%) to 6.71 L (138%), giving a post-bronchodilator FEV_1/FVC ratio of 57%. By plethysmography, total lung capacity was 9.35 L (119%) with residual volume 3.51 L (137%). Diffusion capacity was 19.7 mL·min^{-1}·mmHg^{-1} (57%) and 56% when adjusted for alveolar volume. Repeat CT scan showed radiological emphysema, a fibro-calcific nodule and mild bronchial wall thickening without bronchiectasis. The diagnosis of mild COPD with TBALD was confirmed. Smoking cessation advice was provided along with nicotine replacement therapy. In addition, an inhaled bronchodilator was commenced with olodaterol/tiotropium two inhalations once daily 2.5 µg/2.5 µg.

Following the completion of multidrug-resistant (MDR)-TB treatment with a successful microbiological outcome, PTLD was suspected given the history of multiple TB episodes, smoking and radiological abnormalities on prior imaging. Given the significant smoking history and the noncalcified pulmonary nodules, lung cancer was considered, despite the absence of other clinical features. A repeat CT scan of the chest was obtained. This demonstrated two enlarging pulmonary nodules with stable fibro-calcific changes of PTLD and background radiological emphysema. Subsequent 2-fluoro-2-deoxy-D-glucose (FDG)-positron emission tomography scan showed FDG avidity within both low-density nodules with standardised uptake value of >8 in each nodule. There was no FDG uptake in any of the nonenlarged mediastinal lymph nodes and no evidence of distant metastases in the chest or elsewhere. In addition, a brain CT with contrast showed no evidence of intracranial metastatic disease. A subsequent CT-guided lung core biopsy was obtained from the lingular nodule and this showed high-grade nonsmall cell carcinoma (adenocarcinoma). Following a multidisciplinary lung cancer team discussion, a diagnosis of early-stage nonsmall cell lung cancer of the left upper lobe was made and a presumed, synchronous right upper lobe primary lung cancer was thought likely. Given the patient's age, good lung function, Eastern Cooperative Oncology Group performance status of 1 along with normal echocardiogram and no significant comorbid disease, curative treatment was recommended. The patient underwent left upper lobe resection with curative intent followed by stereotactic radiotherapy for the (presumed) right upper lobe synchronous primary lung cancer (no biopsy was possible).

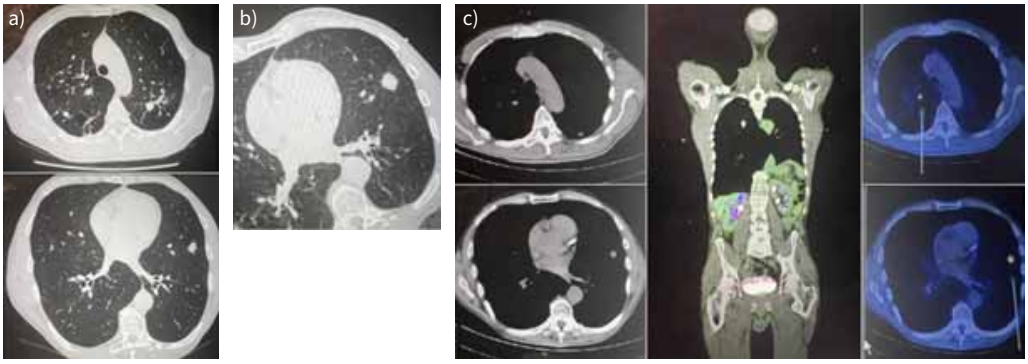


FIGURE 3 A 60-year-old smoker with successfully treated pulmonary multidrug-resistant tuberculosis (TB) with two enlarging pulmonary nodules (right upper lobe and lingula) on computed tomography (CT) scan at TB treatment completion. a) CT at TB diagnosis with emphysema and nodular scarring; b) CT at TB completion with enlarging, lobulated pulmonary nodule; c) 2-fluoro-2-deoxy-D-glucose (FDG) positron emission tomography scan showing FDG-avid nodules in right upper lobe and lingular segment of the left upper lobe.

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one of the recent studies focusing on shortening TB treatment, reported no difference in respiratory health at week 96, based on the mMRC dyspnoea score and the proportion of patients with a forced expiratory volume in 1 s (FEV_1) <50% between patients on the experimental regimens and the standard of care [30].

Risk factors for PTLD

Current knowledge about risk factors for PTLD is limited, and the shortage of evidence, the heterogeneous clinical picture of PTLD, and the nonspecific nature of its definition make its interpretation more challenging. A recent systematic review and meta-analysis of risk factors for PTLD found 64 studies overall, with a number of risk factors reaching statistical significance: older age (OR 1.62, 95% CI 1.07–2.47), previous TB treatment (OR 3.43, 95% CI 2.37–4.97), smoking (OR 1.41, 95% CI 1.09–1.83), alcohol use (OR 1.84, 95% CI 1.04–3.25), smear-positive disease (OR 3.11, 95% CI 1.77–6.44), and radiographic pulmonary lesions at baseline (OR 2.04, 95% CI 1.07–3.87) [31].

It is likely that poor treatment outcomes and pulmonary comorbidities in TB patients predispose to PTLD. Poverty, air pollution and malnutrition are key social determinants of health, and are probably major contributors to the risk of PTLD. Even if evidence is scarce, it is likely that an accumulation of exposures known to damage pulmonary function will worsen PTLD outcomes. Smoking is associated with poorer TB treatment outcomes and delayed culture conversion [32, 33]. Furthermore, smoking, and indoor and outdoor air pollution are associated with chronic lung disease, often with an obstructive pattern. A study from Mozambique showed a risk ratio of 7.5 for persistent lung injury as measured by spirometry at week 52 post-TB treatment initiation in TB patients with a history of >10 pack-years of smoking compared with those without [34]. Diabetes is also associated with a higher risk of poor treatment outcomes and mortality in TB [35]. Even if much of what has been written on PTLD relates to adults, pulmonary sequelae are known to occur in paediatric and adolescent TB survivors [36–38].

Late diagnosis of TB disease is associated with more severe lung damage [39, 40]. This is likely to increase the risk of PTLD. Delayed diagnosis or start of treatment may be due to a combination of health service- or patient-related factors. Asymptomatic or subclinical TB presents a particular challenge, underlining the importance of active TB case finding. Although not consistently reported, patients with asymptomatic TB have less lung destruction as judged from radiological changes. However, the threshold at which patients report symptoms varies between individuals, and in some the extent of disease may have advanced by the time TB is diagnosed.

Multidrug-resistant (MDR)-TB is often associated with more advanced lung impairment and poorer outcomes than drug-susceptible TB. A systematic review of spirometry results post-TB demonstrated a combined estimated mean FEV_1 of 76.6% and forced vital capacity (FVC) of 81.1% predicted for drug-susceptible TB and 65.9% and 76.0%, respectively, for MDR-TB [25]. This may in part be due to a greater likelihood of previous TB in the same individual and also delayed start of treatment for the current episode when access to rapid and comprehensive drug susceptibility testing and appropriate medication is low.

Chronic lung diseases, like COPD, are associated with chronic airflow limitation and exacerbated by noxious inhalants or gases. Very little is known about the role of COPD as a risk factor for PTLD, whereas TB itself is associated with chronic airway obstruction [25]. There is a three-fold risk of COPD in patients who have survived TB compared with those without a TB history, with a stronger association in high TB incidence countries, in those with MDR-TB and in never-smokers [8]. Considering the pathophysiology of COPD, it is likely that patients with underlying COPD have worse PTLD and general health outcomes after pulmonary TB, although the evidence is scanty [41]. Patients with bronchiectasis which developed before or after TB are particularly prone to secondary infections (*e.g. Pseudomonas* spp., *Aspergillus* spp., nontuberculous mycobacteria) [42]. This may aggravate PTLD. Silicosis and silicotuberculosis are associated with high pulmonary morbidity [43].

Among individuals aged ⩾15 years, those living with HIV have been reported to have less severe PTLD compared with those who are HIV-negative [44]. In a recent systematic review and meta-analysis (including 32 studies with 6225 patients), the 16 studies that included people living with HIV registered a much lower prevalence of PTLD in people with HIV than for those without HIV (32.8 *versus* 66.9%) [45]. The diagnosis of PTLD in this review was based on a combination of symptoms, radiological changes and lung function impairment, and results suggest that the risk of PTLD is approximately double in patients without HIV infection. This finding can probably be explained by the reduced host immune response in people living with HIV, in particular those with a low CD4 count [9]. However, as HIV-infection is associated with higher mortality in TB patients, results might have a bias towards survivors.

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Over the years, TB treatment trial outcomes have focused on relapse-free cure and survival by end of treatment. Other long-term outcomes, including general lung health and lung function, persistence of respiratory symptoms, psychosocial impact and quality of life have been given less importance, and data on these aspects thus remain rare [18]. Post-TB lung health remains a research need and consensus is needed on which outcomes should be measured at which time points to improve the comparability of outcomes across studies.

Risk of TB recurrence and reduced life expectancy in PTLD

TB recurrence after successful treatment remains a significant challenge for patients with both drug-susceptible and drug-resistant TB [46]. A comprehensive meta-analysis estimated a recurrence rate of 2.26 per 100 person-years, with ∼70% of TB episodes attributable to relapse rather than re-infection [47]. The median time to pulmonary TB recurrence is ∼18 months post-TB treatment, predominantly due to endogenous relapse, but can vary by setting [48, 49].

Analysis of risk factors shows that early recurrences are often due to relapse, whereas re-infections are associated with HIV, prior incarceration, and infection with specific strains [50]. Social and clinical vulnerabilities, such as substance use, mental health disorders, and marginalised living conditions, also contribute to higher recurrence rates [51]. Currently, no validated immune biomarkers exist to predict TB recurrence. Their future development could substantially aid clinical management.

Multiple studies document an increased risk of death among TB survivors compared to matched controls [11, 52]. There is evidence that people with non-rehabilitated PTLD suffer a worsening of their functional parameters, which is likely to contribute to loss in life expectancy [41]. In Brazil, TB survivors experienced a six-fold increase in mortality (particularly within the first year after diagnosis [53]), while in the Republic of Korea, adjusted analyses revealed a 1.62-fold higher risk [54]. Overall, a meta-analysis reported that TB survivors face nearly a three-fold increased mortality risk, with South African data showing a standardised mortality ratio of 3.8 after successful treatment [55]. Despite adequate therapy, TB survivors lose on average 3.6 years of life compared with those without TB, and the estimated global burden of TB increased by an additional 6.1 million DALY (a 54% increase) when post-TB mortality and morbidity were considered [52].

Respiratory infections, partly stemming from lung impairment, as well as cardiovascular disease contribute significantly to post-TB mortality [12, 25, 53, 56, 57]. Diabetes is particularly associated with cardiovascular mortality [53]. Other known risk factors for reduced life expectancy and poor outcomes such as alcohol and substance use disorders, mental health conditions and homelessness may overlap in the same individual [53, 54]. Conversely, higher body mass index and regular physical activity appear to confer protection [54].

Clinical management of TBALD and PTLD

Recognising the long-term impact of TB on lung health requires not only awareness of its clinical spectrum and risk factors, but also well-structured strategies for assessment and care through person-centred care, including social protection. Given the dearth of trial evidence on how TB treatment influences lung function and quality of life, the management of PTLD in most settings remains unstructured [7]. A practical action checklist is proposed based on clinical experience and practice, although not all elements are supported by formal evidence (table 1). These actions aim to support clinicians in preventing PTLD during and after the end of TB treatment, and managing PTLD should it develop. Timely diagnosis and management of PTLD can minimise further lung impairment and reduce the risk of respiratory insufficiency and infections, while early screening for comorbidities and lifestyle modifications can lower risk of cardiovascular and other diseases [58].



During TB treatment, it is important to diagnose and treat TBALD that are commonly associated with PTLD. For example, bronchiectasis is frequently detectable on chest radiography or CT, and COPD can be confirmed by post-bronchodilator spirometry. Bedside spirometry, which is standard for diagnosis in other respiratory insufficiency and other respiratory diseases including asthma and COPD, may be done after TB treatment completion to screen for lung function abnormalities.

After TB treatment is completed, preventing death from avoidable complications remains relevant. A diagnosis of TB recurrence should be excluded in a TB survivor presenting with productive cough or other respiratory symptoms. In such patients there is an added risk of drug resistance. While nucleic acid amplification tests are helpful to detect *M. tuberculosis* and drug-resistant strains, they may detect nonviable bacilli that persist in the sputum of people who had TB for many months after their disease

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TABLE 1 Checklist of assessment and preventive measures for post-tuberculosis (TB) lung disease

	Domain	Assessment or intervention
During TB treatment	Symptom evaluation	Persistent cough, dyspnoea, chest pain, haemoptysis
	Vital signs and BMI	Monitor weight and height (BMI), pulse rate, respiration rate, oxygen saturation
	Imaging	Chest radiography to assess progression or complications and CT scan if needed
	TB treatment appropriateness	Ensure correct regimen matched with drug susceptibility test results, dosing and monitoring of adverse events
	Respiratory insufficiency and other comorbidities	Identify and manage pre-existing or concurrent conditions
	Smoking status and cessation	Assess smoking (and vaping) history, offer cessation counselling and support
	Environmental exposures	Assess biomass smoke, occupational risks such as exposure to silica dust
	Clinical examination	Respiratory-focused physical examination
	Functional testing (if feasible)	Office-based spirometry or peak flow
	Vaccination status	Review vaccination status for influenza, pneumococcus and SARS-CoV-2
After TB treatment	Symptom reassessment	Persistent or worsening respiratory symptoms
	Imaging	Repeat chest radiography or CT scan if symptoms persist or abnormalities suspected
	Assessment of lung function and respiratory insufficiency	Spirometry post-treatment, D_{LCO} and lung volumes if available
	Cardiac assessment	Echocardiogram, if heart failure or pulmonary hypertension are suspected
	Radiological interpretation	Evaluate for fibrosis, cavitation, bronchiectasis, pleural thickening
	Nutritional status	Reassess BMI and dietary intake
	Functional capacity	Functional testing and walking test, physical exercise
	Comorbidity screening and follow-up	Ongoing management of COPD, asthma, cardiovascular disease, HIV, diabetes and others
	Rehabilitation needs	Refer to pulmonary rehabilitation if applicable
	Smoking cessation follow-up	Reinforce smoking (and vaping) cessation strategies, support relapse prevention
	Psychosocial and palliative care	Assess quality of life, mental health, supportive care needs, alcohol and substance use disorders
	Vaccination	Administer relevant vaccinations and booster doses (including influenza, pneumococcus and SARS-CoV-2)
	TB recurrence or reinfection	Evaluate new symptoms, consider microbiological reassessment

BMI: body mass index; CT: computed tomography; SARS-CoV-2: severe acute respiratory syndrome coronavirus 2; D_{LCO}: diffusing capacity of the lung for carbon monoxide.

resolved [59]. A post-TB survivor with bronchiectasis should be assessed and treated for common organisms associated with structural lung disease, including bacteria (*Haemophilus influenzae*, *Streptococcus pneumoniae*, *Pseudomonas aeruginosa*), fungi (*Aspergillus* spp., *Candida* spp.) and non-TB mycobacteria (*Mycobacterium avium intracellulare*, *Mycobacterium abscessus*). In addition, advice on sputum clearance techniques, nutrition and vaccination are also reasonable interventions in PTLD management for TB survivors.

Smoking cessation is a key intervention, recommended at diagnosis of TB and reinforced throughout treatment and follow-up [60]. Tobacco use is a major contributor to lung function decline and can worsen the impact of TB-associated pulmonary damage [61]. Other environmental lung health insults, such as exposure to biomass smoke, silica and other occupational dusts and air pollutants, should also be assessed and addressed through counselling and support where feasible, while broader public health interventions are implemented. While not all based on evidence-based recommendations, routine care commonly includes monitoring of height, weight and body mass index; identification and management of comorbid respiratory conditions; radiological evaluation; assessment of vital signs including oxygen saturation and, if possible in people who are not infectious, lung function testing.

Vaccination against severe acute respiratory syndrome coronavirus 2, influenza and pneumococcus should be addressed where appropriate at the time of TB diagnosis, or during and after treatment completion [62]. After treatment, follow-up should include evaluation of symptoms and clinical status, repeat imaging when indicated, spirometry and diffusion testing if available, and assessment of rehabilitation needs. When relevant, pulmonary rehabilitation, breathing support, and end-of-life care should be considered [41]. There should also be a focus on social support, social protection and prevention of TB recurrence or reinfection.

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From a patient's perspective, distressing symptoms like persistent cough, haemoptysis, chest pain and breathlessness need attention, regardless of whether they occur during or after TB treatment. In this context, the role of bronchodilators remains uncertain [63]. A person-centred approach encourages health providers to listen to the individual's needs and consider if there might be something else that could be done to improve symptoms or quality of life, utilising existing knowledge and tools for respiratory disease. Once PTLD is suspected or diagnosed, a systematic approach to evaluation is essential to define its severity, guide clinical decisions, and identify opportunities for intervention. Early identification of delayed diagnosis, implementation of infection prevention and control measures, and timely initiation of treatment and social measures are all essential to reduce the burden of PTLD [64].

Finally, education and counselling by health care professionals enables TB survivors to recognise early warning signs of TB recurrence or PTLD, and seek prompt medical attention is essential to reduce avoidable mortality. Patient education should begin during treatment and intensify upon its completion, helping survivors understand the potential complications and the importance of ongoing care.

Discussion and conclusion

The persistent global burden of TB underscores the urgent need to not only prevent and treat TB, but also to address the long-term consequences faced by its survivors due to TBALD and PTLD. Despite significant advancements in diagnostics, treatment and social interventions in recent decades, TB remains a leading cause of chronic disease worldwide, particularly in low- and middle-income countries, where most people affected by TB live. In addition to an increased risk of death and recurrence, many TB survivors face disability, limiting their enjoyment of life, social functioning, and capacity to earn a living through meaningful employment. Recognising the multifaceted nature of TB is key to the design of effective strategies that will address all people in need of care. Ignoring TBALD during the course of treatment, and discharging TB patients definitively from care once they complete their treatment, is thus no longer acceptable. WHO guidance on the collection of data on sustained treatment success after the end of TB medication could direct more programmatic attention to post-treatment follow-up [65].

Microbiological cure and reduced transmission have historically been the main goals of TB treatment, and these remain the principal reported outcomes in research. TBALD and PTLD may have devastating effects on the individual, both medically and socially, despite achieving TB cure [66]. The concept of TBALD and PTLD is gaining increasing attention as access to curative treatment has broadened and more people survive a bout of TB. The clinical diversity of TBALD and PTLD, ranging from asymptomatic to severe respiratory impairment, calls for a systematic clinical assessment and tailored management. Addressing risk factors such as air pollution, smoking, alcohol use, and comorbidities, along with early detection of disease and start of effective treatment, can mitigate long-term impairment and disability, improve health-related quality of life and increase survival. Radiological and functional evaluations are critical to define disease severity, guide appropriate interventions, and monitor progression. However, integrating these assessments into routine care to make them available to the whole population remains a challenge, especially in resource-limited settings, and may impact on health equity.

The risk of TB recurrence and the heightened mortality risk among TB survivors should stimulate renewed discussion about the value of long-term follow-up. Routine screening and structured follow-up for PTLD remain uncommon, leading to missed opportunities for early intervention. Well-functioning primary health systems offer opportunities for a rapid, reliable diagnosis of TB. However, in many settings this does not happen, and people will be unaware of a pre-existing respiratory disease or an impairment to their lung function. For many, especially young people, the diagnosis of pulmonary TB may be the first interaction with healthcare and the first time that they are aware of respiratory pathology.

TB healthcare providers have an important opportunity to diagnose and manage lung conditions from the moment TB is diagnosed. They play a central role in preventing, diagnosing and managing care for people affected by TBALD and PTLD. Providers offer follow-up care, monitor lung function, and treat complications like chronic respiratory issues. They also educate patients about managing their condition and preventing further lung damage. It is important that post-TB care is integrated into health services and strategies. Healthcare providers and national TB programmes will need to consider a longer perspective in continuous care, health education, and social support and social protection for TB survivors and their families. In 2025 WHO released a policy brief stressing the importance for health programmes to address the overlapping needs of people affected by TB and TBALD [67]. WHO is currently preparing guidance on TBALD and PTLD for TB providers, researchers and policy-makers, building upon the integrated approach to lung health for TB patients and survivors.

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In many high-TB-burden countries, TB care is delivered as part of a specific structure of the health ministry: the national TB programmes. These programmes organise preventive and curative services through different arrangements, be it their own health facilities or else *via* integrated primary healthcare services. Importantly, they also reach out to the health services that are responsible for TB care outside their usual span of control, such as the private sector, prisons, occupational services and general hospitals. Multidisciplinary approaches to care that integrate pulmonary rehabilitation, smoking cessation, management of comorbidities, appropriate vaccination and social protection measures are essential.

There is also a need for more and better-quality evidence to strengthen recommendations for TBALD and PTLD in adults, adolescents and children, making a more convincing case for practitioners and policy-makers alike. More research on effective treatments and prevention are needed. Some of the priority research questions relate to our understanding of the interplay between the host immune system and *M. tuberculosis* that predispose to PTLD. In addition, we need more evidence about which TB treatment regimens best avert PTLD and which host-directed therapies can attenuate or prevent PTLD. Exploring ways to integrate pulmonary rehabilitation and long-term follow-up in TB-endemic or low-resource settings is essential for equitable access to care in the world.

Sustained financial support is vital to strengthen healthcare, social support and research. Funding ensures the availability of resources for PTLD management, including medications, rehabilitation programmes, human resources and training for healthcare providers [68]. Health insurance coverage needs to provide for TB and TBALD, and patients helped to avail of healthcare entitlements that they have. Recent global cuts to funding are expected to have a profound impact on efforts to curb TB and make it even more important to sustain the response to TB and TBALD due to their major socioeconomic impact and the high cost-effectiveness of the interventions recommended for their management. PTLD prevention and management need to be recognised as core components of TB strategies.

For this to materialise in the near future, a concerted approach is needed from multiple public health actors to identify those at risk of TB and TBALD and take targeted action. A large proportion of TB cases occur in adults in their productive age. Quantifying the indirect costs of PTLD, such as reduced productivity and lost income, may strengthen advocacy for investment in long-term care. Ultimately, strategies that combine effective prevention, prompt diagnosis, accessible treatment, and comprehensive post-treatment care are vital to reduce TB's long-term health and socioeconomic impact. Ending TB could be a success story for the history of humanity and encourage the world to come together more often to address other threats to global health.

Key points

- Different lung pathologies may precede, accompany and follow tuberculosis (TB), and some may predispose to TB or worsen TB outcomes, such as COPD and silicosis.
- TB-associated lung disease (TBALD) refers to a broad range of respiratory abnormalities, concurrent or attributable at least in part to TB, while post-TB lung disease (PTLD) (a form of TBALD) occurs after successful completion of treatment.
- Clinical features of TBALD range from mild symptoms (with minimal radiological changes) to severe respiratory failure.
- Early detection and integrated care of TBALD can improve outcomes for TB survivors.

Self-evaluation questions

1. Give three reasons why TB is still considered a major, global public health concern today?
2. Give three examples of TB-associated lung diseases.
3. When can action be taken to detect and prevent TB-associated lung disease?
4. What interventions are commonly used to prevent, treat or alleviate illness caused by TB-associated lung disease?

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(ERS) Assembly 10: Respiratory Infections. She also reports travel support from ERS to attend the ERS Spring Meeting in Zürich in 2025. C. Vilaplana declares being the secretary (unpaid) of the Tuberculosis and Non-Tuberculous Mycobacteria Diseases group (10.02) of the ERS Assembly 10: Respiratory Infections and declares complimentary registration to the ERS Conference 2025 for this role. She also declares membership of 2 non-profit foundations: FUITB (www.uitb.cat/fundacio-uitb/) and ACTMON (www.actmon.org/index.php). In addition, she declares grants to her employer from the Catalan Government through 2021 SGR 00920. D. Falzon and K. Viney are staff members of the World Health Organization (WHO). The authors alone are responsible for the views expressed in this article and they do not necessarily represent the views, decisions or policies of the WHO. D. Falzon also reports support from ERS, namely complimentary registration and coverage of travel expenses for participation in the annual ERS congress as a speaker, and other funds to WHO, his employer, for activities on TB. O.W. Akkerman is an associate editor of this journal.

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Suggested answers

1. 1) World's top infectious disease killer with 1.3 million deaths and 11 million new TB cases per year; 2) eminently avoidable and treatable with current tools that are widely available, although inequities persist and TB remains very stigmatising; 3) chronic impairment impacts the health-related quality of life after cure and long-term survival despite successful microbiological treatment.
2. Noncommunicable: COPD, bronchiectasis, pulmonary fibrosis, pulmonary hypertension, lung cancer; communicable: recurrence of TB after past TB, influenza, coronavirus disease 2019, chronic pulmonary aspergillosis.
3. Before TB develops in people at risk of TB infection, progression to disease, or poor TB outcomes; during treatment for TB disease; upon completion of TB treatment.
4. TB preventive treatment (TPT) in those with TB infection and increased risk; early detection and treatment of TB disease to forestall TB sequelae; reduction of other lung insults such as cigarette smoke, occupational dust exposure, other indoor or outdoor air pollution, and vaccination; pulmonary rehabilitation during and after TB treatment for symptomatic relief and to improve functional capacity and confidence; oxygen therapy in those who are hypoxic; ventilation in the most severe cases.

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